5 Chapter 5: Magnetostatics

5.0 Preliminaries

In the previous chapter, we explored the relationship between electric field, $\vec{E}$, and electric flux density, $\vec{D}$, which are mediated via the property of permittivity, ε. For magnetic fields, the subject of this chapter, there is analogous relationship described by

$$\vec{B} = \mu\vec{H},$$

where $\vec{B}$ is the magnetic flux density, $\vec{H}$ is the magnetic field intensity, and μ is the relevant medium's **magnetic permeability**. For materials with constant magnetic permeability μ, the above relationship holds. The difference in magnetic fields versus electric fields is that magnetic fields are only generated by *moving* charges; in other words, current.

5.1 Magnetic Forces and Torques

Charges that *move* in a magnetic field experience a force. Given a point charge, for example, with charge q with velocity $\vec{u}$ in a magnetic flux density $\vec{B}$, then

$$\vec{F}_m = q\vec{u} \times \vec{B} \text{ N},$$

where $\vec{F}_m$ represents the **magnetic force**.

Clearly, if the charge is *stationary*, it will experience no magnetic force, since $\vec{u} = 0 \ \frac{\text{m}}{\text{s}}$. For example: To find $\vec{F}_m$ on a $+q$ moving at a velocity $\vec{u} = \hat{\alpha}_y u_0$ in a magnetic flux density $\vec{B} = \hat{\alpha}_z B_0$.

$$\begin{aligned}
&\vec{F}_m = q\vec{u} \times \vec{B} \\
&\vec{F}_m q(\hat{\alpha}_y u_0 \times \hat{\alpha}_z B_0) \\
&\vec{F}_m = \hat{\alpha}_x q u_0 B_0 \text{ N}.
\end{aligned}$$

Additionally, we must consider the possibility of both magnetic and electric forces on the point charge. The total force on the point charge is simply the sum of the two forces, as described by the Lorents Force equation,

$$\vec{F} = \vec{F}_e + \vec{F}_m = q\vec{E} + q\vec{u} \times \vec{B} = q(\vec{E} + \vec{u} \times \vec{B}) \text{ N}.$$

Since moving point charges experience a force in a magnetic field, we can reasonably assume that current carrying conductors also experience magenetic forces under the same circumstances. Generalizing,

$$\begin{aligned}
&q\vec{u} \to Id\vec{l} = \vec{I}dl \\
&d\vec{F}_m = \vec{I} \times \vec{B}dl \\
&\vec{F}_m = \int d\vec{F}_m = \int_L Id\vec{l} \times \vec{B} = I\int_L d\vec{l} \times \vec{B} = \int_L (\vec{I} \times \vec{B}) \ dl.
\end{aligned}$$

For example: given a semicircular conductor carrying a current I, the closed circuit is exposed to a magnetic field $\vec{B} = \hat{\alpha}_y B_0$, find

a) The magnetic force $\vec{F}_1$ on the straight section of the wire.

   For the straight section (bottom of the semicircle) we can define $d\vec{l} = \hat{\alpha}_x dx$, since the entire section is in the x direction. Then,

$$\begin{aligned}
&\vec{F}1 = I\int d\vec{l} \times \vec{B} = I\int \hat{\alpha}_x dx \times \hat{\alpha}_y B_0 \\
&\vec{F}_1 = IB_0\hat{\alpha}_z \int_0^{2r} dx = \hat{\alpha}_z 2IrB_0 \text{ N}.
\end{aligned}$$

b) The magnetic force $\vec{F_2}$ on the curved section. Based on the cross product $d\vec{l} \times \vec{B}$, we know that the force will be in the $-\hat{\alpha}_z$ direction (right-hand rule). Finding $d\vec{l}$, using the cylindrical coordinate system, $d\vec{l} = rd\phi\hat{\alpha}_\phi$. Then,

$$\vec{F_2} = I \int d\vec{l} \times \vec{B} = I \int rd\phi\hat{\alpha}_\phi \times \hat{\alpha}_y B_0$$
$$\vec{F_2} = IrB_0 \int (-\sin(\phi)\hat{\alpha}_x + \cos(\phi)\hat{\alpha}_y) \times \hat{\alpha}_y \; d\phi$$
$$\vec{F_2} = IrB_0(-\hat{\alpha}_z) \int_0^\pi \sin(\phi) d\phi = -\hat{\alpha}_z 2IrB_0 \text{ N.}$$

Therefore, $\vec{F_1} = -\vec{F_2}$, and there is no net force.

Current carrying loops under the influence of magnetic fields can also *rotate* due to magnetic torque. For example, given a magnetic field in the $x - y$ coordinate system $\vec{B} = \hat{\alpha}_y B_y + \hat{\alpha}_z B_z$. Given a current carrying loop (square) in the $x - y$ plane, what current direction will produce torque under the given magnetic field? Testing all the available options,

$$I_1 = \hat{\alpha}_y I$$
$$I_2 = -\hat{\alpha}_x I$$
$$I_3 = -\hat{\alpha}_y I$$
$$I_4 = \hat{\alpha}_x I.$$

Then, using $\vec{F}_m = \int_L (\vec{I} \times \vec{B}) \; dl$,

$$\vec{F_1} = (\hat{\alpha}_y I \times \hat{\alpha}_z B_z) l_1 = \hat{\alpha}_x I B_z l_1$$
$$\vec{F_2} = (-\hat{\alpha}_x I \times \hat{\alpha}_z B_z) l_2 = \hat{\alpha}_y I B_z l_1$$
$$\vec{F_3} = (-\hat{\alpha}_y I \times \hat{\alpha}_z B_z) l_1 = -\hat{\alpha}_x I B_z l_1$$
$$\vec{F_4} = (\hat{\alpha}_x I \times \hat{\alpha}_z B_z) l_2 = -\hat{\alpha}_y I B_z l_2$$

Therefore, for the normal (B_z) component of the field, the loop is pulled in all directions equally, and no torque results. Finding the effect of the B_y components,

$$\vec{F_1} = (\hat{\alpha}_y I \times \hat{\alpha}_y B_y) l_1 = 0$$
$$\vec{F_2} = (-\hat{\alpha}_x I \times \hat{\alpha}_y B_y) l_2 = -\hat{\alpha}_z I B_y l_1$$
$$\vec{F_3} = (-\hat{\alpha}_y I \times \hat{\alpha}_y B_y) l_1 = 0$$
$$\vec{F_4} = (\hat{\alpha}_x I \times \hat{\alpha}_y B_y) l_2 = \hat{\alpha}_x I B_y l_2.$$

Therefore, net force is still zero, but now, since $\vec{F_1}$ and $\vec{F_2}$ are equal but in opposite directions, there will be a torque that causes rotation in *what direction*?

More generally, torque can be found as $\vec{m} = IA\hat{\alpha}_n$, where $\vec{m}$ is the vector magnetic moment, A is the loop area, and $\hat{\alpha}_n$ is the unit vector normal to the loop (right-hand rule applied to current direction). Then,

$$\vec{T} = \vec{m} \times \vec{B},$$

where $\vec{T}$ is the direction of torque. Alternately,

$$\vec{T} = \hat{\alpha}_T m B \sin(\theta)$$
$$\vec{T} = \hat{\alpha}_T I A B \sin(\theta).$$

5.2 The Biot-Savart Law

The Biot-Savart law defines the static magnetic field produced by a steady current.

To formulate the Biot-Savart law, we take a short section dl of a current carrying wire, and define the resultant magentic field at a point lying at a straight-line distance R from the current carrying wire,

$$d\vec{H} = \frac{I}{4\pi}\frac{d\vec{l} \times \hat{\alpha}_r}{R^2}.$$

Integrating over the entire current carrying wire,

$$\vec{H} = \int d\vec{H} = \frac{I}{4\pi}\int \frac{d\vec{l} \times \hat{\alpha}_r}{R^2},$$

where $\hat{\alpha}_r = \frac{\vec{R}}{|\vec{R}|}$.

Alternately,

$$\vec{H} = \frac{I}{4\pi}\int \frac{d\vec{l} \times \vec{R}}{R^3}\ \frac{\text{A}}{\text{m}}.$$

The direction of the resultant field will be rotating in the ϕ direction around the wire. Note that $d\vec{l}$ is in the direction of the current I.

Solving problems using Biot-Savart:

For example, we have a linear conductor of length l carrying a current of I along the z axis. Find the magnetic flux density $\vec{B}$ at a point P located at a distance of r in the $x - y$ plane.

To make the problem simpler, we may center the conductor at the origin, meaning one end is at $-\frac{l}{2}$ and the other is at $\frac{l}{2}$. We know that the magnetic field will be in the $\hat{\alpha}_\phi$ direction, and we assume a magnetic permeability of μ_0. Therefore,

$$\vec{H} = \frac{I}{4\pi}\int \frac{d\vec{l} \times \vec{R}}{R^3}.$$

First, we need to define a few parameters. In this case, $d\vec{l} = \hat{\alpha}_z dz$, since the conductor exists along the z axis. Finding the difference vector between any point on along the conductor and P, we can express $\vec{R}$ as $\hat{\alpha}_r \cdot r - \hat{\alpha}_z \cdot z$. Then, $d\vec{l} \times \vec{R} = (\hat{\alpha}_z dz) \times (\hat{\alpha}_r \cdot r - \hat{\alpha}_z \cdot z) = \hat{\alpha}_\phi r dz$.

Then,

$$\vec{H} = \frac{I}{4\pi}\int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{\hat{\alpha}_\phi r dz}{(z^2+r^2)^{\frac{3}{2}}} = \hat{\alpha}_\phi \frac{I}{4\pi} \cdot r \int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{dz}{(z^2+r^2)^{\frac{3}{2}}}.$$

Using a standard formula for integration, this reduces to

$$\vec{H} = \hat{\alpha}_\phi \frac{I \cdot r}{4\pi} \cdot \frac{I}{r^2}\left(\frac{z}{\sqrt{r^2+z^2}}\ \Big|_{-\frac{l}{2}}^{\frac{l}{2}}\right) = \hat{\alpha}_\phi \frac{I \cdot l}{2\pi r\sqrt{4r^2+l^2}}.$$

Finding $\vec{B}$,

$$\vec{B} = \mu_0 \vec{H} = \hat{\alpha}_\phi \frac{\mu_0 I \cdot l}{2\pi r\sqrt{4r^2+l^2}}\ \text{T}.$$

We may simplify this equation for an identical conductor to the above conductor, except that it extends to infinity. So, as $l \to \infty$,

$$\vec{B} = \hat{\alpha}_\phi \frac{\mu_0 I}{2\pi r}.$$

Later, we will show that the same problem can be solved more easily using Ampere's Law, the magnetic field analogue to Gauss's Law.

As another example, a circular loop of radius a carries a current of I. Find the magnetic flux density $\vec{B}$ at a point on the axis of the loop.

Use the right-hand rule to determine that, if the current I flows counter-clockwise, the field will be in the $\hat{\alpha}_z$ direction (fingers are current, thumb is field). Since the loop is symmetric (centered at the origin of the $x-y$ plane, the $x-y$ components will cancel, and there will be a net field only in the z direction at any point along the z axis.

Then, finding $d\vec{l}$, we may use cylindrical coordinates to find $d\vec{l} = \hat{\alpha}_\phi a d\phi$. Then, $\vec{R}$ can be expressed as $\vec{R} = \hat{\alpha}_z \cdot z - \hat{\alpha}_r \cdot a$, and $d\vec{l} \times \vec{R} = \hat{\alpha}_r azd\phi + \hat{\alpha}_z a^2 d\phi$.

Simplyfing, $dH \to \hat{\alpha}_z$, and $H = \int d\vec{H}$.

Solving,

$$\vec{B} = \mu_0 \vec{H} = \hat{\alpha}_z \frac{mu_0 I \cdot a^2}{2(z^2+a^2)^{\frac{3}{2}}} \text{ T}.$$

When two current carrying conductors exert magnetic fields forces, there is an analogous relationship to that between two point charges; that is, the superposition principle is such that the forces contributed by each conductor sums with the force exerted by the other. In addition, there is a force on each conductor exerted by the other.

For two parallel conductors with current runnin gin the same direction along the z direction, we may find the force exerted on the second conductor, for example, as

$$\vec{F_2} = (\vec{I} \times \vec{B})L = -\hat{\alpha}_y \frac{\mu_0 I_1 I_2 l}{2\pi d}.$$

Similarly,

$$\vec{F_1} = \hat{\alpha}_y \frac{\mu_0 I_1 I_2}{2\pi d}.$$

The takeaway here is there is an *attractive* force between the wires.

In terms of Maxwell's equations, magnetostatics can be reduced to

$$\nabla \cdot \vec{D} = \rho_v$$
$$\nabla \times \vec{E} = 0$$
$$\nabla \cdot \vec{B} = 0.$$

Moving on, Ampere's Law relates the static magnetic field to the source of the field (current), and states that the line integral of the magnetic field $\vec{H}$ around a closed path is equal to the current flowing through the surface bounded by that path. Mathematically,

$$\oint_C \vec{H} \cdot d\vec{l} = I.$$

In terms of magnetic flux density $\vec{B}$,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I.$$

Using Ampere's Law, we can solve the infinitely long conductor problem from before much more simply.

Using the same conductor, infinitely long and lying along the z axis, we may find the magnetic field intensity at a distance r in the $x-y$ plane. First, we can assume that the magnetic field is in the ϕ direction, so $\vec{H} = \hat{\alpha}_\phi H$. Then, $d\vec{l} = \hat{\alpha}_\phi r d\phi$. Therefore, Ampere's Law in this case reduces to

$$\int_0^{2\pi} \hat{\alpha}_\phi H \cdot \hat{\alpha}_\phi r d\phi = H \cdot r \cdot 2\pi = I.$$

This agrees with our prior assessment, that

$$H = \frac{I}{2\pi r}.$$

As another example, a long straight wire of radius a carries a current I that is uniformly distributed over its cross section. Find the magnetic field $\vec{H}$ a) inside the wire and b) outside the wire.

Again, we may center the wire at the origin of the $x - y$ axis to solve more easily for $\vec{H}$. Applying Ampere's Law again, if we create a closed loop inside the wire with radius $r \leq a$, $H2\pi r = \frac{I}{\pi a^2} \cdot \pi r^2$, using the current density. Simplifying,

$$\vec{H} = \hat{\alpha}_\phi \frac{I \cdot r}{2\pi a^2} \ \frac{\text{A}}{\text{m}}.$$

Outside the loop,

$$\vec{H} = \hat{\alpha}_\phi \frac{I}{2\pi r}.$$

Continuing applications of Ampere's Law, find the magnetic flux density inside a closely wound toroidal coil with N turns of coil carrying a current I. Also find $\vec{B}$ for $r < a$ and $r > b$.

Beginning with $r < a$, we may use Ampere's Law again,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I.$$

Using a closed loop, $B \cdot 2\pi r = \mu_0 \cdot I$. Since $I = 0$; that is, there is no current flowing inside the toroid's inner radius, there is no magnetic field in the region $r < a$, and $\vec{B} = 0$.

Moving on to $a < r < b$, we may once again assume that $B \cdot 2\pi r = \mu_0 \cdot I$. What is the total current in the region $a < r < b$? Based on the number of turns, $I_{\text{total}} = I \cdot N$. Therefore, $\vec{B} = \hat{\alpha}_\phi \ \frac{\mu_0 N I}{2\pi r}$.

Finally, outside the toroid, in the region $r > b$, $I = 0$ and $\vec{B} = 0$.

Note: Double-check Ampere's Law and how it works, doesn't seem to be by *enclosing* current, like Gauss's.

Revisiting Maxwell's equations,

$$\begin{aligned}
\nabla \cdot \vec{D} &= \rho_v \\
\nabla \times \vec{E} &= 0 \\
\nabla \cdot \vec{B} &= 0 \\
\nabla \times \vec{H} &= \vec{J}.
\end{aligned}$$

The last equation correspond's to Ampere's Law, with a slightly different formulation. Integral of current density is current, duh.

Note: Revisit Stokes' Theorem, and relationship to current/current density.

Static magnetic fields, like static electric fields, have boundary conditions. As a review of boundary conditions, we'd like to know how the magnetic field $\vec{H}$ and the magnetic flux density $\vec{B}$ change across a boundary with a difference in magnetic permeability.

The relevant equations are

$$\begin{aligned}
&B_{1n} = B_{2n} \\
&\mu_1 H_{1n} = \mu_2 H_{2n} \\
&\hat{n} \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \\
&H_{1t} - H_{2t} = J_s
\end{aligned}$$

For most cases, $J_s = 0$ and so $H_{1t} = H_{2t}$.

5.3 Inductors and Inductance

Defining an **inductor**, an inductor is an energy storage device that stores energy in a magnetic field (similar to a capcitor, which stores energy in an electric field). As a reminder,

$$C = \frac{Q}{V}.$$

For inductance,

$$L = \frac{\Lambda}{I},$$

that is, flux linkage in webers over current. The energy stored in an inductor may be described as

$$W_m = \frac{1}{2}LI^2.$$

What is flux linkage Λ? Flux linkage of an inductor is the total magnetic flux that links the current; similar to magnetic flux, but if you have N turns, like in the toroid example, the total magnetic flux Φ that passes through those windings is the flux linkage.

Mathematically,

$$\Phi = \int\int_S \vec{B} \cdot d\vec{s}.$$

For N turns,

$$\Lambda = N \cdot \Phi.$$

Common types of inductors:

Solenoids have cylindrical shaped frames and have N turns of current-carrying wire wound around them. Deriving their inductance,

$$L = \frac{\mu N^2 \pi a^2}{l},$$

where a is the radius and l is the length.

Toroids have donut-shaped frames with N turns of current-carrying wire wound around them. Deriving their inductance,

$$L = \frac{\mu N^2 a^2}{2 \cdot r_0},$$

where r_0 is the average? radius and a is??

For example, if the magnetic flux Φ produced by a given current links that same current, then the resulting inductance is self-inductance, but if it links the current in another circuit, then the resulting inductance is mutual inductance (requires two separate circuits/structures).

Self-inductance:

$$L = \frac{\Lambda}{I} = \frac{\mu}{I}\int\int_s \vec{B} \cdot d\vec{s}.$$

Mutual inductance:

$$L_{12} = \frac{\Lambda_{12}}{I_1} = \frac{N_2}{I_1}\int\int_{s_2} \vec{B}_1 \cdot d\vec{s}_{s_2}.$$

The procedure for determining inductance is as follows:

1. Choose a proper coordinate system
2. Assume a current I in the conducting wire
3. Find $\vec{B}$ from I. (i.e., use Ampere's Law)

4. Find Φ (magnetic flux) as

$$\Phi = \int \int_S \vec{B} \cdot d\vec{s}$$

5. Find flux linkage as

$$\Lambda = N \cdot \Phi$$

6. Finally,

$$L = \frac{\Lambda}{I}.$$

As an example, develop an expression for the inductance per unit length of a coaxial line with inner and outer conductors of radii a and b and an insulating material of μ.

Using Ampere's Law, we can find the field that exists in the region $a < r < b$ as

$$\vec{B} = \hat{\alpha}_\phi \frac{\mu I}{2\pi r}.$$

Calculating the flux,

$$\Phi = \int \int_s \vec{B} \cdot d\vec{s} = \int_0^l \int_a^b \hat{\alpha}_\phi \frac{\mu I}{2\pi r} \cdot \hat{\alpha}_\phi \; dr \; dz.$$

Therefore,

$$\Phi = \frac{\mu I}{2\pi} \cdot l \int_a^b \frac{1}{r} \; dr = \frac{uIl}{2\pi} \ln{(\frac{b}{a})}.$$

Finally,

$$\Lambda = N \cdot \Phi = \Phi,$$

and

$$L = \frac{\Lambda}{I} = \frac{\mu l}{2\pi} \ln{(\frac{b}{a})}.$$

Therefore, self-inductance in this case is a function of permeability and inner/outer radius, primarily, if we divide by l to find inductance per unit length.

Mathematically, for a coaxial cable,

$$L' = \frac{\mu}{2\pi} \ln{(\frac{b}{a})}.$$

As a final example, find the mututal inductance between a conducting rectangular loop and very long, straight wire.

Assuming a current of I on the straight wire, we will first find $\vec{B}$ using Ampere's Law in the form $\vec{B} = \hat{\alpha}_\phi \frac{\mu I}{2\pi r}$. Then, derive Φ (**Note:** Review this in the book, brain cells fried).